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UTTAR PRADESH

DECLARATION

I do hereby declare that the report entitled “**PRECISION NITROGEN MANAGEMENT IN RICE AND ESTIMATION OF CROP-WISE AND STATE-WISE ORGANIC MANURE PRODUCTION IN INDIA**” is a manifestation of my research work done and studies conducted by me with several books, online resources, study materials available online, video lectures, research papers, talk shows with scientists, researchers, SRFs and scholars.

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CERTIFICATE

On the basis of the declaration submitted by **Mr. D LOKESH GOUD S/o Mr. D LAXMAN GOUD**, a student of B.Sc. (H) Agriculture (A014119118102), I hereby certify that the project titled “**Precision nitrogen management in rice and estimation of crop-wise and state-wise organic manure production in India**”, which is submitted to **Amity Institute of Organic Agriculture, Amity University, Uttar Pradesh, Noida**, in partial fulfillment of the requirement for the award of the degree of **B.Sc. (H) Agriculture**, is an original contribution to existing knowledge and a faithful record of the work carried out by him under my guidance and supervision.

To the best of my knowledge, this work has not been submitted in part or in full for any degree to this university or elsewhere.

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Precision Nitrogen Management in Rice & Estimation of Crop-wise and State-wise Organic Manure Production in India

1. Introduction

Nitrogen (N) is the most important essential element for the overall growth and development of plants. Despite its high abundance in the air (around 79%), it is not readily available to them; rather, it is added mostly as inorganic fertilizer to agricultural lands (Hakeem *et al.*, 2011). Globally, urea is the most preferred form of nitrogen in agriculture (Modolo *et al.*, 2015). However, once applied to the soil, it undergoes microbially mediated transformations such as hydrolysis, nitrification and denitrification, which are the three main pathways of N loss (Rosmarina *et al.*, 2016). Such losses have economic and environmental implications. Fertilizer N is most often supplied in three chemical forms: ammonium (NH_4^+), nitrate (NO_3^-) and urea.

Nitrification and denitrification are key processes contributing to N_2O emissions from the soil. Ammonia oxidation and successive nitrifier denitrification (the reduction of nitrite (NO_2^-) by ammonia-oxidizing bacteria) or denitrification, among other processes, can be important pathways for N_2O emissions (Guo *et al.*, 2018). In the ammonia oxidation process, N_2O is produced by the chemical decomposition of hydroxylamine (NH_2OH).

One potential way to mitigate N_2O emissions is to use nitrogen-efficient technologies such as nitrification and urease inhibitors, slow-release N fertilizers, the LCC, the SPAD meter and the Green Seeker, together with traditional practices like brown manuring and INM, to slow down the rate of nitrification and reduce the availability of the substrate (NH_4^+) for nitrification in the rice crop.

2. Nitrification Inhibitors (NIs)

Nitrification inhibitors (NIs) slow down the microbial conversion of ammonium-nitrogen to nitrate-nitrogen (nitrification). This reduces the risk of loss through denitrification or leaching and thereby increases the nitrogen use efficiency (NUE) of nitrogen fertilizers (Norton and Ouyang, 2019).

2.1 Synthetic Nitrification Inhibitors

Synthetic NIs such as nitrapyrin reduce nitrate leaching and N_2O emissions, and improve NUE, crop yields and nitrogen uptake (Woodward *et al.*, 2021). Dicyandiamide (DCD) is effective in suppressing nitrification in many agro-ecosystems. It effectively inhibited gross N nitrification in paddy soils (gross N includes all residual emissions of environmentally harmful nitrogen compounds from agriculture into the soil, water and air), increased grain yields by 19% and N-recovery efficiency by 30–40% over urea, and reduced N_2O emissions by 73% (TingLan *et al.*, 2013; Gaihre *et al.*, 2020). Dimethylpyrazole phosphate (DMPP) is also effective in increasing soil NH_4^+ -N content when combined with urea, organic and inorganic fertilizers (Yang *et al.*, 2016), and it can lower soil N_2O emissions in temperate environments.

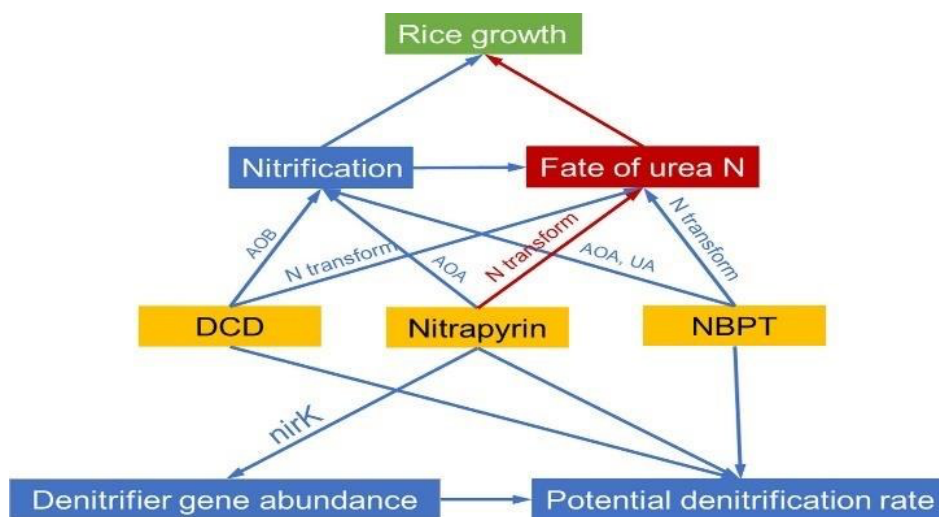


Fig. 1: Conceptual model for the effect of nitrification inhibitors on rice growth.

The use of NIs is a technique that can improve N fertilizer use efficiency, reduce nitrate leaching and reduce the emission of the climate-relevant gas N_2O . The first enzyme involved in the oxidation of NH_4^+ to NO_3^- in soils is ammonia monooxygenase (AMO). NIs inhibit AMO, which directly decreases the nitrification rate and reduces the NO_3^- concentration that serves as the substrate for denitrification. In this way, the two main pathways of N_2O production in soils are blocked. Although archaea have also been shown to oxidize NH_3 , the most important target for NIs is the enzymatic activity of NH_3 -oxidizing bacteria, because it is much more strongly affected (Ruser and Schulz, 2015).

Meta-analyses report that the application of urease inhibitors and NIs significantly reduced inorganic N leaching (48%), N_2O emission (44%) and NO emission (24%) (Burzaco *et al.*, 2014;

Qiao *et al.*, 2015; Thapa *et al.*, 2016), while increasing crop yield (7.5%) and NUE (12.9%) (Abalos *et al.*, 2014). The synthetic NIs commonly used in agricultural practice are listed in Table 1.

Table 1: Common Synthetic Nitrification Inhibitors.

N-Source	Base Compound	N-Process	Common Names	N-Content	Inhibition Duration (weeks)
Nitrapyrin	2-chloro-6-trichloromethylpyridine	Nitrification, denitrification	N-Serve, Stay-N 2000	–	2–6
DCD	Dicyandiamide	Nitrification	DCD, Ensan	1.6	4–8
DMPP	3,4-dimethylpyrazole phosphate	Nitrification	Entec, DMPP	12–26	6–8

Source: Tisdle *et al.* (2014)

NIs in admixture with urea may enhance losses through ammonia volatilization under certain soil situations, for example when surface-applied or on bare soils. However, these effects need to be investigated in the presence of growing crop plants and established crop canopies (Sahrawat, 1980).

2.2 Natural Nitrification Inhibitors

Botanical (natural) NIs encapsulate conventional soluble fertilizer materials and provide a protective cover that controls the entry of water and the rate of dissolution. This makes nutrient release gradual and increases availability in step with plant requirements (Fujita *et al.*, 1992). They also improve soil health by reducing nitrification and N₂O emission, and they enhance crop productivity (Banik *et al.*, 2016).

Natural inhibitors like neem cake improved the recovery efficiency of applied N in an arable soil very effectively. Neem oil has been found to inhibit the nitrification rate by 20–50% in soil, slightly lower than synthetic NIs such as DCD (56–80%) (Hala *et al.*, 2014; Raza *et al.*, 2019). Another potential natural NI is karanj (*Pongamia pinnata*) seed extract, which minimizes N₂O emission from soil (Banik *et al.*, 2016). It acts as a highly efficient NI (62–75% reduction in nitrification) as well as an N₂O mitigator (92–96% reduction in N₂O emission) (Majumdar, 2002).

Mahua extracts containing saponins could also be responsible for slow mineralization of N. The seed cake and extracts contain saponins, which are responsible for nitrification inhibition. In an

incubation study using clay loam soil, the inhibitory effect of mahua cake extract persisted for only 20 days (Kumar *et al.*, 2015).

NIs inhibit the microbial enzymes responsible for the conversion of NH_4^+ to NO_3^- . Stopping the nitrification process improves the nitrogen recovery rate and agronomic nitrogen use efficiency where N losses are very high. Several chemical inhibitors are still at the experimental level; high cost, limited availability, adverse effects on beneficial soil microorganisms and, above all, poor extension services and product promotion are major constraints. It is therefore necessary to develop plant-based NIs for improving nitrogen use efficiency and crop productivity, and for environmental safety. Natural NIs are easily available, cheap, pro-nature, farmer-centric and eco-friendly (Upadhyay *et al.*, 2011). Some natural NIs obtained from different plant parts are listed in Table 2.

Table 2: Natural Nitrification Inhibitors (NNI).

Common Name	Scientific Name	Active Compounds	Reference
Neem	<i>Azadirachta indica</i>	Azadirachtin	Slangen and Kerkhoff (1984)
Karanj	<i>Pongamia glabra</i>	Karanjin, glabrin, glabrosaponin	Modolo <i>et al.</i> (2015)
Mahua	<i>Madhuca longifolia</i>	Saponin	Bisht <i>et al.</i> (2018)

NIs act on *Nitrosomonas* and *Nitrobacter* bacteria, which carry out the conversion of ammonium to nitrite and of nitrite to nitrate, respectively. The core characteristics of an effective NI are listed below (Slangen and Kerkhoff, 1984).

- **Specificity:** The NI should block the conversion of ammonium to nitrate (the activity of *Nitrosomonas*) and be non-toxic to other soil organisms, humans and animals.
- **Persistence:** The material should stay active in the soil for the required period. Compounds that degrade rapidly will not be useful.
- **Mobility:** The NI should move with the fertilizer and nutrient solution. Compounds with very high vapour pressure move too fast, and easily adsorbed compounds are not effective.
- **Economy:** The chemical should be cheap.

2.3 Ideal Conditions Where Use of NI is Recommended

- Use of NIs may be advantageous where losses of N through leaching and denitrification are accompanied by nitrification of fertilizer nitrogen.
- NIs are more effective in light-textured soils, since the effectiveness of these compounds decreases quickly in heavy-textured soils.
- NIs should be applied to the soil microsites where nitrification occurs, rather than to the entire soil volume, so that their concentration remains high enough for a reasonable period.

3. Urease Inhibitors (UIs)

3.1 Synthetic Urease Inhibitors

These chemical compounds block urease enzyme activity. Urease is found in soil as well as in plant residues; together with water, it hydrolyzes (breaks down) urea into ammonium. The loss process that UIs protect against is ammonia volatilization. In soils with high pH, or in soil–residue environments that are poorly buffered against a rise in pH, rapid hydrolysis of urea results in an accumulation of ammonia (NH_3) rather than ammonium.

UIs such as N-(n-butyl) thiophosphoric triamide (NBPT) significantly inhibit urea hydrolysis, indirectly affecting nitrification. NBPT decreased the potential denitrification rate (PDR) at the rice heading stage but had little effect on denitrifier gene abundance, unlike nitrapyrin, which decreased *nirK* gene abundance (Meng *et al.*, 2020). Urea treated with NBPT reduces NH_3 loss by around 53%, and the yield gain averages 6.0%, varying from 0.8 to 10.2% depending on the crop species (Cantarella *et al.*, 2018).

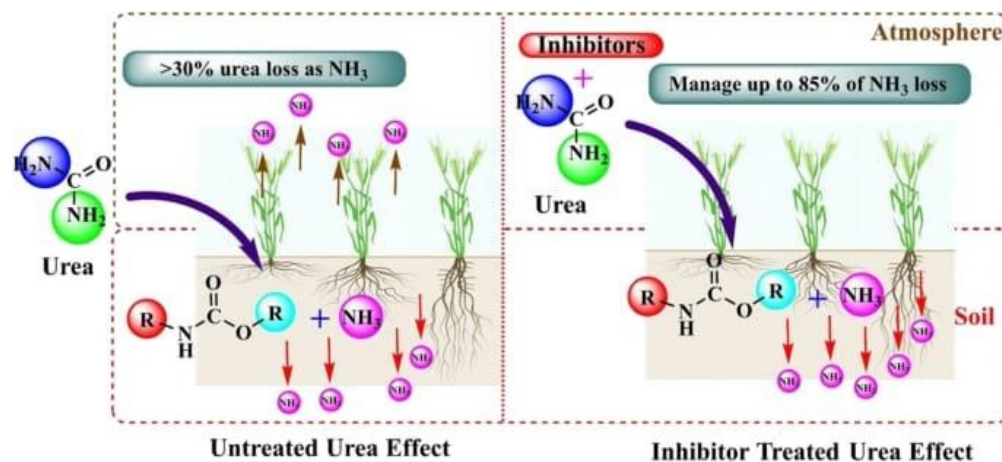


Fig. 2: Effect of untreated urea and inhibitor-treated urea.

Thiosulphates are UIs used to reduce ammonia volatilization from urea or urea ammonium nitrate (UAN) fertilizer applied in crop fields. Urea containing ammonium thiosulphate (ATS) reduced NH_3 volatilization losses by up to 11% compared with UAN alone (Sloan and Anderson, 1995). ATS alone or in association with urea did not affect the soil microbial biomass pool. On the other hand, a field experiment on Canadian clay loam and fine sandy loam soils showed inconsistent results for urease inhibition by ATS (Modolo *et al.*, 2018).

The inhibitor N-(n-propyl) thiophosphoric triamide (NPPT) has shown similar benefits in reducing ammonia volatilization losses after urea application. Application of an NBPT + NPPT mixture reduced NH_3 volatilization losses to 0–6% of applied N, compared with losses of up to 25% without it (Li *et al.*, 2015; Hull, 2018; Hardke, 2020).

Hydroquinone (HQ) significantly reduced NO emissions by around 5% (Modolo *et al.*, 2018). Application of 12 kg ha^{-1} HQ on an alluvial soil, in conjunction with $120 \text{ kg urea-N ha}^{-1}$, decreased N_2O emission by 5% in rice and 7% in wheat compared with crops grown with $120 \text{ kg urea-N ha}^{-1}$ alone (Modolo *et al.*, 2018). The reduction in N_2O emission when NIs and UIs were applied with urea ranged from 5% with hydroquinone to 31% with thiosulphate in rice. Hydroquinone and thiosulphate increased emissions compared with urea alone; however, the global warming potential (GWP) was lower with the inhibitors (except hydroquinone) than with urea alone (Malla *et al.*, 2005).

The properties of various synthetic UIs are presented in Table 3, and crop responses to their application with urea are presented in Table 4.

Table 3: Properties of Synthetic Urease Inhibitors.

Source	Common Names	Base Compound	N-Process	N Content (%)	Inhibition Duration (weeks)
NBPT	Agrotain	N-(n-butyl) thiophosphoric triamide	Volatilization	46	2 to 3
Thiosulphate	ATS, CaTS	Ammonium or calcium thiosulphate	Volatilization, nitrification	12	2 to 3
NPPT	Limus	N-(n-propyl) thiophosphoric triamide	Volatilization	—	—
Hydroquinone	HQ	Hydroquinone	Volatilization	—	—

Source: Tisdle *et al.* (2014)

Table 4: Selected studies on crop and N responses after application of urea-based fertilizers with urease inhibitors.

Crop Type	Crop Response	Soil Nitrogen Response	Urease Inhibitor	Reference
Rice (<i>Oryza sativa</i>)	Magnitude of the effectiveness of inhibitors was associated with soil, season, climate and cultivation system	Slower and reduced losses by ammonia volatilization; slowed and delayed conversion of N to NH ₃ in no-till system; reduced volatilization losses under no-till and conventional tillage	NBPT	Marchesan <i>et al.</i> (2013)
Rice (varieties Apo and Hanyou3, a drought-tolerant hybrid)	Improved plant growth and seed germination (from 9.32 to 16.22% for Apo and from 17.76 to 36.81% for Hanyou3)	Reduced ammonia volatilization	NBPT	Qi <i>et al.</i> (2012)

3.2 Natural Urease Inhibitors

Natural urease inhibitors are obtained from plant parts. These compounds block urease enzyme activity and thus reduce N losses. Allicin, a plant-based bio-inhibitor from garlic (*Allium sativum* L.), can stop urease activity and shows inhibitory effects on soil urease, making it suitable for agricultural applications. Its inhibition was about 75% lower than that of NBPT at steady state. It has the potential to lower the loss of urea in soil and can be used along with urea for improved utilization of applied N by plants (Mathialagan *et al.*, 2017; Beshr Khateeb *et al.*, 2016).

Polyphenolic extracts obtained from the bark of *Acacia decurrens* (green wattle; Fabaceae) or from the seed coat of *Terminalia chebula* (inknut; Combretaceae) inhibited both soil urease and pure urease to the same extent as mercuric chloride and catechol, which are known urease inhibitors. With urea–polyphenol mixtures, NH₃ volatilization from the soil surface decreased after fertilization, demonstrating the potential of tannin-like polyphenols from green wattle and inknut as potent urease inhibitors (Fernando and Roberts, 1976). The natural urease inhibitors are listed in Table 5.

Table 5: Natural Urease Inhibitors.

Inhibitor	Obtained From	Reference
Allicin	Garlic (<i>Allium sativum</i>)	Matczuk and Siczek (2021)
Tannin (polyphenolics)	<i>Acacia decurrens</i> (green wattle)	Modolo <i>et al.</i> (2015)
Quercetin	<i>Allium cepa</i>	Modolo <i>et al.</i> (2015)

UIs gradually slow down urea hydrolysis for a period of 7 to 14 days by suppressing urease activity. During this time, surface-applied urea can move into the soil profile, lowering the concentration of NH_4^+ at the soil surface. UIs like NBPT can reduce ammonia volatilization and nitrite (NO_2^-) accumulation in soil and influence the kinetic and thermodynamic behaviour of soil urease. The efficiency of inhibitors depends mainly on soil temperature and pH and on the concentration of the inhibitor. Most inhibitors, including NBPT, are highly effective in neutral soils with a small range of organic matter. In addition to chemical inhibitors, some natural products such as phenolic compounds (methyl gallate, stilbenoids and flavonoids) can suppress urease activity (Hussain *et al.*, 2021).

4. Slow/Controlled Release Nitrogen Fertilizers

These are granulated fertilizers that differ from regular fertilizers by releasing nutrients slowly or gradually into the soil. They contain a form of plant nutrient whose availability for plant uptake is delayed, or extended for longer than a reference rapidly-available fertilizer such as ammonium nitrate, urea or ammonium phosphate. The delay in initial availability, or the extended period of continued availability, may occur through a variety of mechanisms, including control of water solubility by semi-permeable coatings (Varadachari and Goertz, 2010).

4.1 Types of Slow/Controlled Release Nitrogen Fertilizers

(a) Slow-release N fertilizers: These are among the alternatives for minimizing nitrogen losses from urea and improving its uptake by crops. They consist of fertilizer granules whose surface is covered by hydrophobic chemicals that provide a physical barrier against water, promoting the gradual release of urea into the soil solution (Akiyama *et al.*, 2010). Because they release their nutrients over time, N is always available. These fertilizers are water-insoluble, meaning they

dissolve slowly, which increases NUE and decreases nutrient loss. Some slow-release N fertilizers are listed in Table 6.

Table 6: Slow-release N fertilizers (organic-N low-solubility compounds).

N-Source	Base Compound	Common Names	N Content (%)	Inhibition Duration (weeks)
Urea formaldehyde	Ureaforms, methylol urea	Nitamin, Nitroform, Folocorn	35–40	6–10
Isobutylidene diurea	Isobutylidene urea	IBDU	31	10–16
Triazone	Triazone/urea	N-Sure	28–33	6–10
Melamine	2,4,6-triamino-1,3,5-triazine	Nitrazine	50–60	6–12
Crotonylidene diurea	Urea crotonaldehyde	Crotodur, Triabon	34	6–12

Source: Tisdle *et al.* (2014)

(b) Coated N fertilizers: These are slightly soluble in the soil solution, and their N release rate depends on microbial activity and hydrolysis. Polymer-coated fertilizers are the most recent technology for controlling N release and reducing N losses through leaching, denitrification and volatilization. Some commonly used coated N fertilizers are listed in Table 7.

Table 7: Coated N fertilizers.

N-Source	Base Compound	Common Names	N Content (%)	Inhibition Duration (weeks)
Neem-coated urea	Urea	NCU, NICU (Nimin-coated urea)	46	2–6
Polymer-sulphur-coated urea	Urea	Polyplus, Poly-S	38–42	6–16
Sulphur-coated urea	Urea	Enspan, SCU	30–42	4–12
Polymer-resin-coated urea	Urea	Polygon, Meister, Escote	38–44	8–14

Source: Tisdle *et al.* (2014)

(c) Polymer-coated multi-nutrient fertilizers: These supply all three major fertilizer elements (N, P and K) required for plant growth and development. They release nutrients gradually over extended periods, ranging from as short as 3 months to as long as 18 months. Examples include Osmocote, Multicote and Nutricote.

5. Brown Manuring

Brown manuring is a method of growing the paddy crop and *Sesbania* together. When the *Sesbania* plants overtake the paddy crop in height, at about 25 days after sowing, the herbicide 2,4-D is applied to kill them. Four to five days after spraying, the *Sesbania* plants turn brown and start dying. Their leaves fall to the ground and form a mulch, which increases soil fertility as it decomposes and also helps smother weeds. 2,4-D is a selective herbicide: it kills only the *Sesbania* plants and not the rice plants. This is called the knock-down effect.

Sesbania acts first as a live cover that interferes with weeds, and later, as a dead residue mulch, adds organic matter to the soil (post-killing period). Brown manure crops are grown between the rows of the main crop, which increases plant density and leaves no free space for weeds to grow, resulting in minimal or no weed population. After application of 2,4-D, the *Sesbania* crop decomposes and releases its nutrients faster than with in-situ incorporation. *Sesbania* adds C and N to the soil, which favours microbial activity (Phukan and Bora, 2012).

Other leguminous green manure crops such as cowpea, sunhemp, dhaincha and lentil are also potential brown manure crops, and kharif pulses are especially suitable. The nutrient content and carbon–nitrogen (C:N) ratio of green manure crops are given in Table 8, and the effect of brown manuring on soil organic carbon and post-harvest available nitrogen is given in Table 9.

Table 8: Nutrient content and C:N ratio of major green manure crops.

Crop	Scientific Name	Total N	C:N Ratio	Total P	Total K
Sunhemp	<i>Crotalaria juncea</i>	3.97	21:1	0.37	4.80
Dhaincha	<i>Sesbania aculeata</i>	1.90	44:1	0.34	3.60
Sesbania	<i>Sesbania speciosa</i>	2.71	40:1	0.53	2.21

Source: Iliger *et al.* (2017)

Table 9: Effect of brown manuring on organic carbon content of soil and post-harvest available nitrogen.

Year	Initial soil organic carbon (%)	Organic carbon after harvest (%)	Increase in organic carbon (%)	Initial soil available N (kg/ha)	Soil available N after harvest (kg/ha)	Increase in soil available N (%)
2014	0.54	0.69	0.15	283.0	320.2	13.7
Mean	0.56	0.70	0.14	284.19	322.4	13.4

Source: Samant and Patra (2016)

5.1 Advantages of Brown Manuring

- Competes with weeds, thus reducing their growth.
- Reduces the nitrogen requirement of the crop, as legumes fix atmospheric nitrogen through the bacteria in their root nodules.
- Prevents the loss of water through evaporation and thus helps in water conservation.
- Reduces the cost of cultivation by lowering weed control costs and the fertilizer N requirement.
- Increases soil organic carbon content and soil fertility.

5.2 Green Manure vs Brown Manure

The differences between green manure and brown manure are given in Table 10.

Table 10: Green Manures vs Brown Manures.

Green Manures	Brown Manures
Moisture is necessary for incorporation and decomposition.	Moisture is conserved during the practice.
There is a risk of soil surface erosion.	The plants are left standing, which protects lighter soils from the risk of erosion.
A microbial population is necessary for decomposition.	Chemical desiccation takes place.
The manure crop is incorporated by tillage before seed set, usually around flowering.	It is a no-till version of green manuring, where herbicides are used to kill the manure crop and weeds.

Source: Patil *et al.* (2020)

6. Leaf Colour Chart (LCC)

The Leaf Colour Chart (LCC) is used to determine the N fertilizer needs of the rice crop by measuring the greenness of the rice leaf, which indicates its N content. The LCC has replaced the SPAD chlorophyll meter for estimating leaf nitrogen status and timing N fertilizer applications among farmers. It is a simple, inexpensive and easy-to-use tool with colour shades ranging from yellowish-green to dark green (four-panel and six-panel versions are in use).

Readings should be taken once a week using the LCC, starting 14 days after transplanting (DAT) for transplanted rice and 21 days after seeding (DAS) for wet-seeded rice. Readings are taken

every 7–10 days until the first flowering stage. The critical value is 3 for varieties with light-green foliage and 4 for all other varieties and hybrids. When the leaf colour drops below the critical value, 35 kg N/ha during the kar/kuruvai/navarai season and 30 kg N/ha during the samba/thaladi/pishanam season should be applied.

Because the N supply matches the crop demand, LCC-based N management gives optimal yield increases, and the optimum N supply saves nitrogenous fertilizer to the tune of 20–40 kg/ha (Sudhalakshmi *et al.*, 2008).

6.1 Merits and Demerits of the LCC Tool

The LCC is used for precise management of N fertilizer and to reduce N fertilizer wastage. Its merits and demerits are discussed in Table 11.

Table 11: Merits and Demerits of the LCC Tool.

Merits of LCC	Demerits of LCC
The LCC is an uncomplicated and effortless tool for farmers to measure the nitrogen status of the leaf and to identify the right time for top dressing of N in paddy.	The LCC fails to detect minor variations in leaf greenness when the colour lies between two shades.
The LCC is cheap and portable, making it easy to carry to the field for estimating the N status of the leaf.	The relative accuracy of the LCC in measuring leaf N status can be judged only when it is compared and correlated with chlorophyll meter readings and calibrated for the plant groups.
It is a non-destructive method and does not involve any laboratory analysis.	The LCC is used only to adjust top-dressed N; it cannot be used to decide basal N application.
No specific knowledge or skill is required, because using the LCC involves only comparing the colour of the leaf with a standard chart.	The LCC is better suited to a site-specific nutrient management approach, in which other nutrients must not be limiting to realize the optimal response to N fertilizer. Hence, sufficient levels of other nutrients need to be applied on the basis of soil test results.

Source: Bhupenchandra *et al.* (2021)

6.2 LCC Observation and Drawbacks



Fig. 3: Recording LCC observations in the ICAR-IIRR paddy field.

The topmost fully expanded leaf of each hill was selected, and its colour was compared by placing the middle part of the leaf on the LCC. Nitrogen was applied as per the treatment whenever the colour of more than 5 out of 10 leaves was equal to or below the set critical LCC score. The average LCC reading was determined for each treatment. Readings were taken in the morning (8–10 AM) under the shade of the body to avoid the influence of sunlight, which can alter the perceived LCC colour.

Drawbacks of the LCC:

1. If measurements are not taken in shade, sunlight influences the readings.
2. Deficiencies can be identified only after symptoms have developed, by which time the crop may already have been affected.
3. An LCC developed for a particular place may not be applicable to other regions, and it may not suit hybrid rice and HYVs.
4. The LCC cannot give exact values like laboratory analysis.

7. Green Seeker

The Green Seeker is a hand-held optical reflectance sensor that uses active radiation in the red and near-infrared bands, independent of solar conditions.



Fig. 4: Green Seeker device showing a reading.

The Green Seeker is an integrated optical sensing and application system that measures crop status in response to the crop's nitrogen requirement, so that N can be applied in the right amount, at the right place and at the right time in real time (4R stewardship). The device directly outputs the NDVI (normalized difference vegetation index) and RVI (ratio vegetation index), generating readings at a rate of 10 per second.

The sensor unit has self-contained illumination in both the red band (656 nm, with about 25 nm full width at half maximum, FWHM) and the near-infrared band (774 nm, with about 25 nm FWHM). The sensor samples at a very high rate (approximately 1,000 measurements per second) and averages the measurements between outputs. It outputs NDVI at 10 readings per second at a slow walking speed of about 0.5 m s^{-1} .

Sensor readings (NDVI and RVI) were collected 0.5 m above the rice canopy across each plot, excluding plot borders, and the average values were used to represent each plot. Readings were collected at five growth stages: tillering, panicle initiation, booting, before heading and heading.

The original technology was developed for large farms; however, a small hand-held version costing a fraction of the original (approximately Rs. 40,000) is now commercially available. A simple, farmer-friendly Android application, Padi2U, has been developed using the Green Seeker NDVI value to precisely compute the amount of urea for top dressing, increasing N use efficiency.



Fig. 5: A field expert collecting data using the Green Seeker device.

8. Integrated Nutrient Management (INM)

INM is the judicious use of all possible nutrient sources to meet the plant nutrient requirement at an optimum level, sustaining the desired crop productivity with minimal impact on the environment. In INM, the immediate nutrient requirement of the crop is met through chemical fertilizers, so the rate and time of chemical fertilizer application should be synchronized with the real-time need of the crop. The slow, long-term release of nutrients from organic sources meets the long-term needs of the crop.

The goals of INM are:

- (i) To optimize the benefits from all possible sources of plant nutrients in an integrated manner to achieve a given level of crop production.
- (ii) To maintain the nutrient-supplying capacity of the soil for sustainable crop productivity, ensuring higher nutrient use efficiency, minimal nutrient loss and mitigation of harmful environmental impacts.
- (iii) To minimize the use of chemical fertilizers, thereby reducing the cost of cultivation and enhancing profitability.

8.1 Components of INM

- **Organic manures:** Farmyard manure, compost, vermicompost, biogas slurry, poultry manure, crop residues and bio-wastes such as press mud and sugarcane bagasse.
- **Green manures and green leaf manures:** Dhaincha (*Sesbania aculeata*), *Sesbania rostrata*, sunhemp (*Crotalaria juncea*), *Pongamia glabra*, *Leucaena leucocephala* and *Azadirachta indica*, as well as crop rotation with legumes.

- **Chemical fertilizers:** Urea, ammonium sulphate, ammonium nitrate, calcium ammonium nitrate (CAN), etc.

8.2 INM in Rice

Different types of nitrogen application under INM in rice are given in Table 12.

Table 12: Different types of nitrogen application in INM.

S. No.	Recommendation	Yield	Reference
1	100% of the recommended dose of N through green manure, with 50% NPK	62.7 q ha ⁻¹	Bhandari <i>et al.</i> (1992)
2	50% of N through green manure and the remainder through chemical fertilizers	7.3 t ha ⁻¹	Sharma and Subehia (2014)
3	Soil-based BGA biofertilizer at 10 kg ha ⁻¹ along with 90 kg urea	10% higher yield	Mohanty <i>et al.</i> (2019)



Fig. 6: Graphical representation of INM statistics in rice.

9. Time of Application of Nitrogen Fertilizer

Correct timing of fertilizer application is essential for good growth and development of the rice crop. Rice requires N in large amounts, and the recommended dose of nutrients (RDN) is advised to be broadcast in three splits over the growing season: one-third before planting, incorporated into dry soil; one-third at mid-tillering; and one-third at panicle initiation. The effect of different times and methods of N application on the rice crop is presented in Table 13.

Table 13: Time of application of fertilizer for the rice crop.

S. No.	Method	Times of Application	Yield	Reference
1	Direct-seeded upland rice	3 splits: 1/2 at 20 days, 1/4 at tillering, 1/4 at panicle initiation	More than expected	Sharma <i>et al.</i> (2007)
2	Direct-sown rice under lowland conditions	4 splits: 17% at 21 DAS, 33% at 35 DAS, 33% at panicle initiation, 17% at first flowering	4.18 t ha ⁻¹	Thilagavathi and Ramanathan (2005)
3	Direct-sown rice under lowland conditions	4 splits: 1/6 at 15 DAS, 1/3 at tillering, 1/3 at panicle initiation, 1/6 at flowering	4.92 t ha ⁻¹	Sathiya and Ramesh (2009)
4	Aerobic rice	4 splits: 1/6 at 15 DAS, 1/3 at tillering, 1/3 at PI, 1/6 at flowering (recorded higher tillers)	2.82 t ha ⁻¹	Sathiya and Ramesh (2009)

10. Soil Plant Analysis Development (SPAD) Meter

The Soil Plant Analysis Development (SPAD) meter, or chlorophyll meter, developed by the Minolta Company, is a simple, portable diagnostic tool that measures leaf greenness or relative chlorophyll content (Dominika and Matczuk, 2016). It is mainly used to identify crop N status and relative chlorophyll content.

It enables users to quickly and easily measure potential photosynthetic activity, which is closely linked to leaf chlorophyll content, crop nitrogen status and leaf greenness. SPAD readings are strongly influenced by the part of the leaf where measurements are made, because chlorophyll is not evenly distributed along the leaf blade. The meter measures the difference between the transmittance of red (650 nm) and infrared (940 nm) light through the leaf, generating a three-digit SPAD value (Uddling *et al.*, 2007). Samples do not need to be sent to a laboratory, saving time and money.

Several factors, such as growth stage, cultivar, specific leaf weight, leaf thickness, leaf position on the plant, measurement location on the leaf, environmental stress and solar radiation, can significantly affect chlorophyll meter readings (Zhao *et al.*, 2016). Leaf area-based N concentration (N_a) has a unique linear relationship with the SPAD values of rice plants at all growth stages (Peng *et al.*, 1995). SPAD readings indicate plant N status, and the amount of N to

be applied is determined by the physiological N requirement. Critical SPAD values for the rice crop in different seasons are given in Table 14.

Table 14: Critical SPAD values for different seasons, cropping conditions and rice varieties.

Crop Establishment	Varietal Group	Panicle Density (per m ²)	Critical SPAD Value	
			Dry Season	Wet Season
Transplanted rice	Traditional improved local aromatic rice	300–400	30–32	30
	Semi-dwarf indica varieties	400–500	32–35	35–37
	Hybrid rice	400–500	32–35	35–37
Wet direct-seeded rice				
Broadcast sown	All varieties	High (~800)	29–30	30
		Medium (400–500)	32	35
Row seeded (drum seeded)	All varieties	High (600–650)	32	32
		Medium (400–500)	32–35	32–35

Note: Values are for Philippine conditions. Source: Balasubramanian *et al.* (2000)

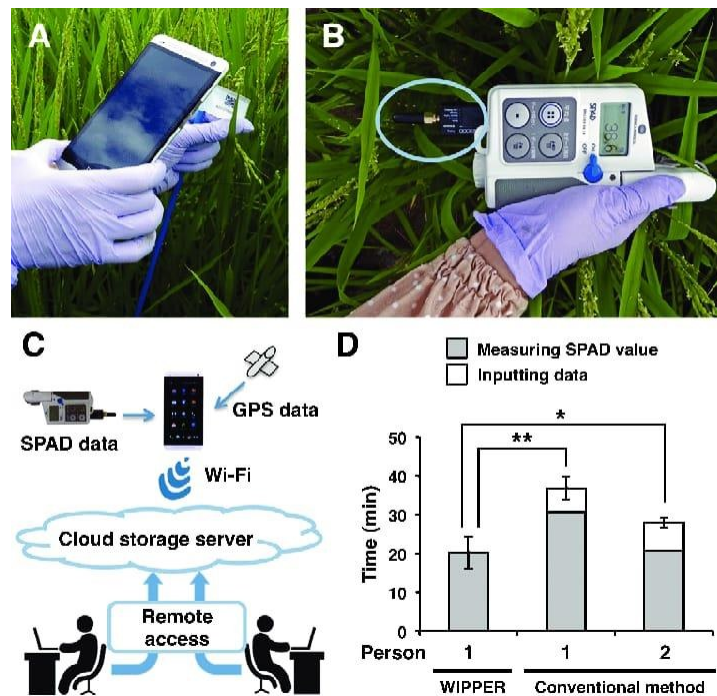


Fig. 7: SPAD meter readings and data collection.

10.1 Cost and Utility of the Chlorophyll Meter

The chlorophyll meter is too costly for a small-scale farmer to buy: it costs around Rs. 1 lakh. It is therefore mainly used by agronomists to assess the nitrogen status of the rice crop.

To identify the N requirement of rice more affordably, scientists developed the Leaf Colour Chart (LCC), which is low-cost and easy to understand. With the LCC, farmers can easily identify the amount of nitrogen required by the crop.

11. Estimation of Crop-wise and State-wise Organic Manure Production in India

11.1 Organic Manure Production from Cereals & Millets (2019-20)

Crop	Organic Manure Production (2019-20)
Bajra	62.1 lakh tonnes
Barley	395 lakh tonnes
Coarse cereals	286.4 lakh tonnes
Jowar	2863 thousand tonnes
Maize	172.5 lakh tonnes
Ragi	10.5 lakh tonnes
Rice	713.2 lakh tonnes
Small millets	2.2 lakh tonnes
Wheat	679.5 lakh tonnes

11.2 Organic Manure Production from Oilseeds (2019-20)

Crop	Organic Manure Production (2019-20)
Castor	1289.4 thousand tonnes
Groundnut	7464 thousand tonnes
Linseed	84.4 thousand tonnes
Niger	28.9 thousand tonnes
Rapeseed & mustard	6386.5 thousand tonnes
Safflower	30.5 thousand tonnes
Sesamum	4602.7 lakh tonnes
Soybean	959.7 lakh tonnes
Sunflower	0.63 lakh tonnes

11.3 Organic Manure Production from Pulses (2019-20)

Crop	Organic Manure Production (2019-20)
Black gram	531.7 million tonnes
Gram	3234.9 million tonnes
Green gram	641 million tonnes
Horse gram	75.8 million tonnes
Khesari	47.6 million tonnes
Lentil	281.8 million tonnes
Moth	81.9 million tonnes
Peas & beans	219.7 million tonnes
Tur	994.3 million tonnes

11.4 Organic Manure Production from Animal Husbandry (2019-20)

Livestock	Organic Manure Production (2019-20)
Cattle	2366912.5 million tonnes
Buffalo	90215.9 million tonnes
Sheep	35746.6 million tonnes
Goat	67928.6 million tonnes
Pig	15990 million tonnes
Yak	158 million tonnes
Horses & ponies	1675 million tonnes
Donkey	451 million tonnes
Camel	1021 million tonnes
Poultry	36898.6 million tonnes

12. Conclusion

Rice requires more nitrogen fertilizer than most other crops, and nitrogen emissions from agriculture are higher than those from any other fertilizer nutrient. We must therefore meet the crop's requirement by supplying nitrogen fertilizer judiciously, while also protecting the environment from hazardous N emissions that lead to severe climate change impacts.

To tackle these emissions and increase N use efficiency, we need to adopt precision nitrogen management technologies such as nitrification and urease inhibitors, the Green Seeker, the SPAD meter and the LCC, together with cultural methods like brown manuring and INM. Such sustainable paddy cultivation ensures a safer environment and sustainable, nature-centric yield production, which is the need of the hour in the present-day scenario.

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